

(A) Observable layers

Preference (P)

stated categories &
preferred sources
weighted profile state

Logged exposure (E)

ranked recommendation lists
user-date exposure sets

Consumption (C)

clicked articles
click distributions

App-surface pathways (S): conditioning context

e.g., headline · category · home/feed · newsroom · article-detail · search · notifications

(B) Relation-specific audits

RQ1

P-C alignment

- top-category match
- category Jaccard
- weighted cosine

RQ2

E-C traceability

- same-day hit
- six-month recommendation-history overlap
- top-k lift

RQ3

E-C diversity shift

- count matching
- entropy / HHI / top-share gaps
- JS divergence

RQ4

Structured divergence

- bounded diagnostic model
- dense-history checks
- future-window diagnostic

(C) Observability boundaries

Logged exposure

≠ total exposure

Click consumption

≠ complete attention

Current profile snapshot

≠ versioned preference history

Platform traces

remain partial observations